

Part 6

Processes & System Monitoring

Big Idea First (Mental Model)

A **process** is:

| A running program in memory

Every app, service, or command you start becomes a **process**.

Linux gives you tools to:

- **View** processes
- **Monitor** system health
- **Control / stop** processes

SECTION 1: What Is a Process?

Example:

```
firefox
```

- You run Firefox
- Linux creates a process
- That process has a **PID** (Process ID)

Think:

| Program = file
| Process = running instance

SECTION 2: Viewing Processes (**ps**)

Basic view

```
ps
```

Shows:

- Processes in current terminal

Full system view (Very common)

```
ps aux
```

Breakdown:

Letter	Meaning
<code>a</code>	All users
<code>u</code>	User-oriented
<code>x</code>	Background processes

This is a **standard admin command**.

Filter processes

```
ps aux | grep ssh
```

Shows only SSH-related processes.

SECTION 3: Live Monitoring (`top`)

```
top
```

What you see:

- CPU usage
- Memory usage
- Running processes (live)

Key columns:

Column	Meaning
PID	Process ID
%CPU	CPU usage
%MEM	Memory usage

Controls:

- **q** → quit
- **k** → kill process
- **M** → sort by memory
- **P** → sort by CPU

Better version (if available)

htop

- Colorful
- Easier navigation
- Tree view

Often installed manually.

SECTION 4: Killing Processes (**kill**)

Step 1: Find PID

```
ps aux | grep chrome
```

Step 2: Kill it

```
kill PID
```

Example:

```
kill 1234
```

Force kill (use carefully)

```
kill -9 PID
```

Signal	Meaning
15	Graceful stop (default)
9	Force kill

Always try normal kill first.

SECTION 5: Background & Foreground Jobs

Run in background

```
command &
```

Example:

```
sleep 100 &
```

View jobs

```
jobs
```

Bring back to foreground

```
fg %1
```

SECTION 6: System Resource Commands

CPU & memory

```
free -h
```

Shows:

- Used memory
- Free memory

Disk usage

```
df -h
```

Filesystem usage.

Folder size

```
du -sh folder/
```

SECTION 7: Uptime & Load

```
uptime
```

Shows:

- System running time
- Load average

Load average:

| How busy the system is.

SECTION 8: Real-World Professional Scenarios

Find heavy process

```
top
```

Kill frozen app

```
ps aux | grep app  
kill PID
```

Monitor server health

```
top  
df -h  
free -h
```

SECTION 9: Common Beginner Mistakes

Killing wrong PID

Always using `kill -9`

Ignoring system load

Professional habit:

| Observe → Identify → Act